

		Total KE is the same for the same T.
12	A	$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \rightarrow c_{rms} = \sqrt{\frac{3kT}{m}}$ $\frac{c'_{rms}}{c_{rms}} = \sqrt{\frac{T'}{T}} \rightarrow \frac{0.8c_{rms}}{c_{rms}} = \sqrt{\frac{T'}{295}}$ $T = 188.8 \text{ K} = -84.4 \text{ }^\circ\text{C}$
12	A	Total heat supplied by the potatoes = heat for calorimeter + heat for water